Chenhao Zhou

Ph.D. Candidate | Instructor of Record
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Education

Rutgers Business School, Rutgers University *Expected May 2027*
Ph.D. in Management | Concentration: Supply Chain Management | GPA: 4.00/4.00
Advisors: David Dreyfus • Lei Lei (Early-Stage)
Doctoral Committee: Weiwei Chen, Xin Ding, Keith Skowronski

Tandon School of Engineering, New York University *2021*
M.S. in Engineering / Industrial Management

Questrom School of Business, Boston University *2019*
B.S. in Business Administration

Research & Teaching Interests

Research Areas	Teaching Areas
Healthcare operations; workforce management	Operations and supply chain management
Buyer-supplier relationships; innovation	Demand planning and fulfillment
Public policy; environmental sustainability	Supply chain analytics
AI placement in service processes	Service operations; capacity planning
Causal inference methods	Prompt engineering; language model applications

Courses taught: Demand Planning and Fulfillment; Supply Chain Analytics.
Most recent evaluations: Teaching effectiveness 4.78/5; course quality 4.67/5.

Honors, Fellowships, and Research Funding

Summer Research Grant ($4,000 annually, four years)	$16,000	2022–2025
Alfred J. Battaglia Memorial Fellowship ($3,500, awarded twice)	$7,000	2024–2025
Graduate School Academic Excellence Award	$1,000	2025

Personal fellowships and awards to date: **$24,000**.

Funded Project Participation

"Workflow Optimization and Structure Benchmark"	$105,767	2022–2023

Graduate researcher, Rutgers Cancer Institute modeling and simulation team (X. Ding and W. Chen, Co-PIs).
Supported the oncology outpatient workflow study published in *Annals of Operations Research* [1].

Technical Skills

Programming	Python, R, SQL, Stata, MATLAB, LaTeX
Modeling software	AnyLogic, AMPL, CPLEX, Gurobi, OptQuest
Econometrics	Panel data, IV/control functions, local projections, structural estimation
Machine learning	Causal ML; deep reinforcement learning
Visualization	Power BI, Tableau, HTML, React, JavaScript
Stata packages	Curvilinear Local Projections (`clp`): clp-stata.pages.dev
(developer)	U/inverted-U tests (`udiag`): udiag-stata.pages.dev

Publications

[1] **Zhou, C.***, Ding, X., Chen, W., Lei, L., Norrell, J., Tray, A., & Evens, A.M. (2026). "Enhancing efficiency and workflow in oncology outpatient services through simulation-based optimization." *Annals of Operations Research*, 362(1), 457–482. doi.org/10.1007/s10479-026-07181-2

[2] **Zhou, C.** (2021). "House price prediction using polynomial regression with Particle Swarm Optimization." *J. Phys.: Conf. Ser.*, 1802(3), 032034. doi.org/10.1088/1742-6596/1802/3/032034. (Proceedings)

[3] **Zhou, C.** (2020). "Quantitative investment strategy analysis based on machine learning for share dealing." *ICISCE*, 1051–1057. doi.org/10.1109/ICISCE50968.2020.00214. (Proceedings)

Manuscripts in Advanced Review and Revision

[4] **Zhou, C.***, Lin, T., Ding, X., & Chen, W. "Balancing Workforce Fissuring and Service Quality: Evidence from Dialysis Operations." **Under Review (3rd round),** *J. Oper. Manag.* (SI). Last submitted: July.

[5] **Zhou, C.**, Dreyfus, D.*, & Bagchi, A. "An Empirical Examination of Clinician Empowerment: The Role of Resources and Health Disparities on Quality Outcomes." **Under Review (3rd round),** *J. Oper. Manag*. Last submitted: September.

[6] Dreyfus, D., **Zhou, C.***, & Lin, T. "Enhancing Clinical Quality of Care in Socially Disadvantaged Communities: Chain Affiliation and Dynamic Capabilities." **Under Review (2nd round),** *Health Care Manag. Sci.* (SI).

[7] **Zhou, C.***, Li, W., & Alshalfan, A. "When the Fix Is Only Halfway: Sustainable Waste Routing and Toxic Releases." Revision (Reject and Resubmit), *Prod. Oper. Manag.* Deadline: July 2027.

[8] **Zhou, C.***, & Li, W. "When the Choice Is Removed: Leased Owner-Operator Reliance, Mandated Reclassification, and Motor Carrier Safety Performance." *J. Bus. Logist.* (AI-Integrated). Stage 1 proposal accepted; full manuscript invited. Due: February 2027.

Manuscripts Under Initial Review and in Submission

[9] **Zhou, C.***, & Li, W. "The Misplaced Intelligence: AI Position and Output in Sequential Care Lines." Under Review (1st round), *J. Oper. Manag.*

[10] **Zhou, C.***, & Li, W. "When Stars Fall: Quality Instability, Consumer Choice, and the Demand for Long-Term Care." Under Review, *Decision Sciences*.

[11] **Zhou, C.***, Li, W., Dreyfus, D., & Wang, P. "Customer-Base Turnover and Firm Innovation under Trade-Policy Turbulence." *Prod. Oper. Manag.* Editorial encouragement.

[12] **Zhou, C.***, & Li, W. "When Today's Performance Becomes Tomorrow's Benchmark: Dynamic Incentives in Healthcare Value-Based Purchasing." *Manag. Sci.*

[13] **Zhou, C.*** "Supplier Replacement and Lasting Innovation: Curvilinear Local Projections for Recurring Operational Decisions." *J. Oper. Manag.*

[14] **Zhou, C.*** "Beyond the Vertex: Causal Inference for Operating Decisions with U- and Inverted U-Shaped Relationships." *J. Oper. Manag.*

Working Papers

[15] **Zhou, C.*** "Adaptive Online Scheduling for Chemotherapy Infusion Centers: A Deep Reinforcement Learning Approach." Target: *Prod. Oper. Manag.*

[16] **Zhou, C.** "Knowledge Reuse and the Renewal of Innovation." Target: *Manag. Sci.*

[17] **Zhou, C.** "Directional Constraints and the Evolution of Innovation." Target: *Inf. Syst. Res.*

[18] **Zhou, C.**, Li, W.*, & Rogers, D. "Extension Services, Market Remoteness, and Agricultural Productivity: Evidence from Ethiopian Smallholder Farms."

*Corresponding author. **Full papers and submission updates:** www.chenhaozhou.me/research.html

Teaching Experience

Rutgers Business School at Newark and New Brunswick

2025	29:799:310	Demand Planning & Fulfillment	Newark
2026	29:799:310	Demand Planning & Fulfillment	Newark
	33:799:310	Demand Planning & Fulfillment	New Brunswick

Rutgers School of Business–Camden

2026	52:620:326	Supply Chain Analytics	Camden

Course evaluations (affiliated): 2025: 4.91/5, 5.00/5, 85%; 2026*: 4.78/5, 4.67/5, 52%.
*Most recent term; Fall semester in progress.

Teaching Assistant (course sections)

Graduate

- SC Finance (2024; 2)
- Global Proc. & Supply (2023–2024; 2)
- SC Analytics (2023; 1)
- Operations Analysis (2023; 1)
- Project Mgmt. (2022; 1)

Undergraduate

- Demand Planning & Fulfill. (2021–2025; 16)
- SC Analytics Essentials (2023; 2)
- Prod. & Ops. Mgmt. (2024; 1)

Teaching Materials and Supplements

Course Development & Pedagogical Innovation

- Business Analytics & ML Course Platform: modular curriculum, 10+ modules, 3 capstone projects, and Python-based labs.
- Authored AI and coding guides: prompt engineering, LLM applications, Python, Stata, R, and SQL.

Interactive Teaching Materials | www.chenhaozhou.me/teaching.html

Operations & Management Analytics

- M/M/c Queue Model
- Port Management
- Hospital Infusion Center Scheduling (AnyLogic)
- Newsvendor Model
- Vehicle Routing (TSP)
- Facility Location

Supply Chain Finance

- Cash Conversion Cycle
- Reverse Factoring
- Dynamic Discounting
- Portfolio Optimization
- VaR & Risk Metrics
- EOQ Interactive Demo

Handbooks & Student Guides

- Data-Driven Analysis; ML for Business Analytics; Supply Chain Finance; Teaching with Claude AI.
- Endogeneity and Causal Inference Methods: Introduction and Recent Applications (open access).
- Online coding references: Python; Stata; R; SQL.
- Student research: healthcare staffing and research-based teaching simulations.

Certification Examination Support

Lean Six Sigma Green Belt Technical Certificate (Rutgers Business School)	2022–2025
Advanced Certification in Procurement (Rutgers Business School)	2025

Databases and Research Infrastructure

- **Diagnostic-AI adoption panel** — hospital-year adoption and process position of diagnostic AI in coronary care pathways, constructed from Medicare provider-level billing records linked to hospital cost reports. Data paper in preparation. Target: *Manuf. Serv. Oper. Manag.*
- **Buyer-supplier relationships and innovation** — buyer-disclosed suppliers and suppliers' major-customer disclosures extracted from SEC filings, matched to firm identifiers with manual audit and linked to patent records. Data paper in preparation. Target: *Manuf. Serv. Oper. Manag.*
- **CMS healthcare facility panels** — dialysis facilities, nursing homes and skilled nursing facilities, and hospitals; linked records on staffing, ownership, operations, and quality.
- **Additional sectoral data** — product recalls, motor-carrier safety and driver classification, airline-route operations, clinical scheduling, and special economic zones.

Professional and Industry Experience

Cancer Institute of New Jersey, NJ 2022–2024
Research Collaborator (Project-Based)

- Led simulation modeling in a grant-funded partnership with clinical teams.
- Developed agent-based models achieving a 30% reduction in wait time; adopted for deployment.
- Advancing to Phase II: deep reinforcement learning for patient-centered scheduling.

Lujiazui Intl Trust Co., Ltd., Shanghai 2020
Investment Research Associate (Project-Based)

- Evaluated trust-fund and real-estate investments using data-driven forecasting.
- Built ML and PSO-based forecasting models; contributed to two publications (IEEE; J. Phys. 2021).

China Securities Co., Ltd., Beijing 2016
Summer Data Analyst

Academic Service

Conference Presentations

- MSOM Annual Conference, Harvard Business School (2026): 2 papers selected among 36 presentations; both delivered July 2026.
- DSI Annual Meeting (2026): 2 papers accepted to the full-paper track.
- POMS Annual: "Community Health Disparities & Dialysis Performance" (2026).
- POMS Annual: "Team Continuity Amid Workforce Fissure" (2026).
- DSI Doctoral Research Showcase: "Healthcare Operating in a Fissured Workforce" (2025).
- DSI Healthcare Mgmt: "Clinician Empowerment in Resource-Constrained Dialysis" (2025).
- DSI SCM: "Organizational Learning in a Fissured Healthcare Workforce" (2025).
- INFORMS Service Science: "Resilience in Health Care Operations" (2025).
- DSI Healthcare Mgmt: "Clinician Empowerment and the Triple Aim" (2024).
- INFORMS Annual: "Optimizing Oncology Outpatient Flow" (2024).
- INFORMS Annual: "ED Crowding and Diversion Operations" (2023).

Invited Talks & Session Leadership

- Invited Speaker, CHOM Synapse: Healthcare Operations Engagement (2023).
- Session Chair, POMS Healthcare Analytics (2025, 2026); INFORMS Service Science (2025).

Ad hoc Review Service

- Ad hoc reviewer: *Journal of Business Ethics*; *Annals of Operations Research*; *Health Care Management Science*; *Technology and Health Care*.
- Ad hoc conference reviewer: AOM Annual Meeting; DSI — Project Mgmt and Gig Economy (2025), Healthcare (2024).

Professional Memberships

AOM | DSI | IEEE | INFORMS | POMS

References

David Dreyfus, Ph.D.
Associate Professor; Director, MS in Healthcare Analytics, Rutgers Business School
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Dissertation Advisor (Dissertation Committee Chair)

Weiwei Chen, Ph.D.
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Dissertation Committee Member

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Lei Lei, Ph.D.[*]
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Early-stage Advisor (Dissertation Proposal Committee Chair)

Xin Ding, Ph.D.
Dean's Research Professor; Interim Vice Chair, Department of Supply Chain Management, Rutgers Business School
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Dissertation Committee Member

[*] *Dean Lei's later involvement was limited by administrative responsibilities as dean of Rutgers Business School.*